GHAZI UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE

PROGRAMMING FUNDAMENTALS

LAB #04

LABORATORY ASSIGNMENT

STUDENT NAME

Saba Ashraf

ROLL NUMBER

24

PROGRAM

BS Computer Science

SECTION

A

SUBMITTED TO

Sir Asad Majeed

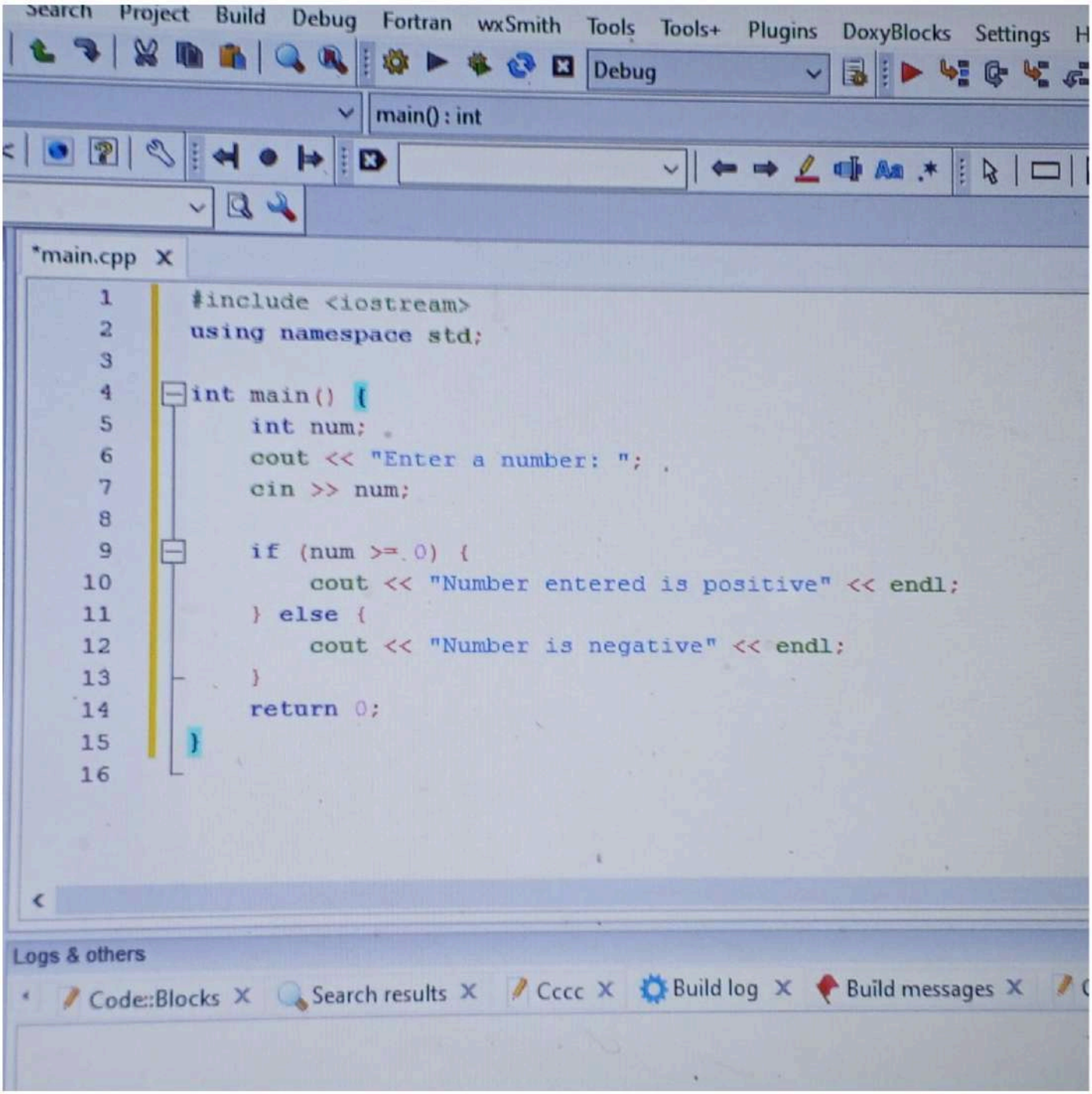
Lab - 04 - Task- 01

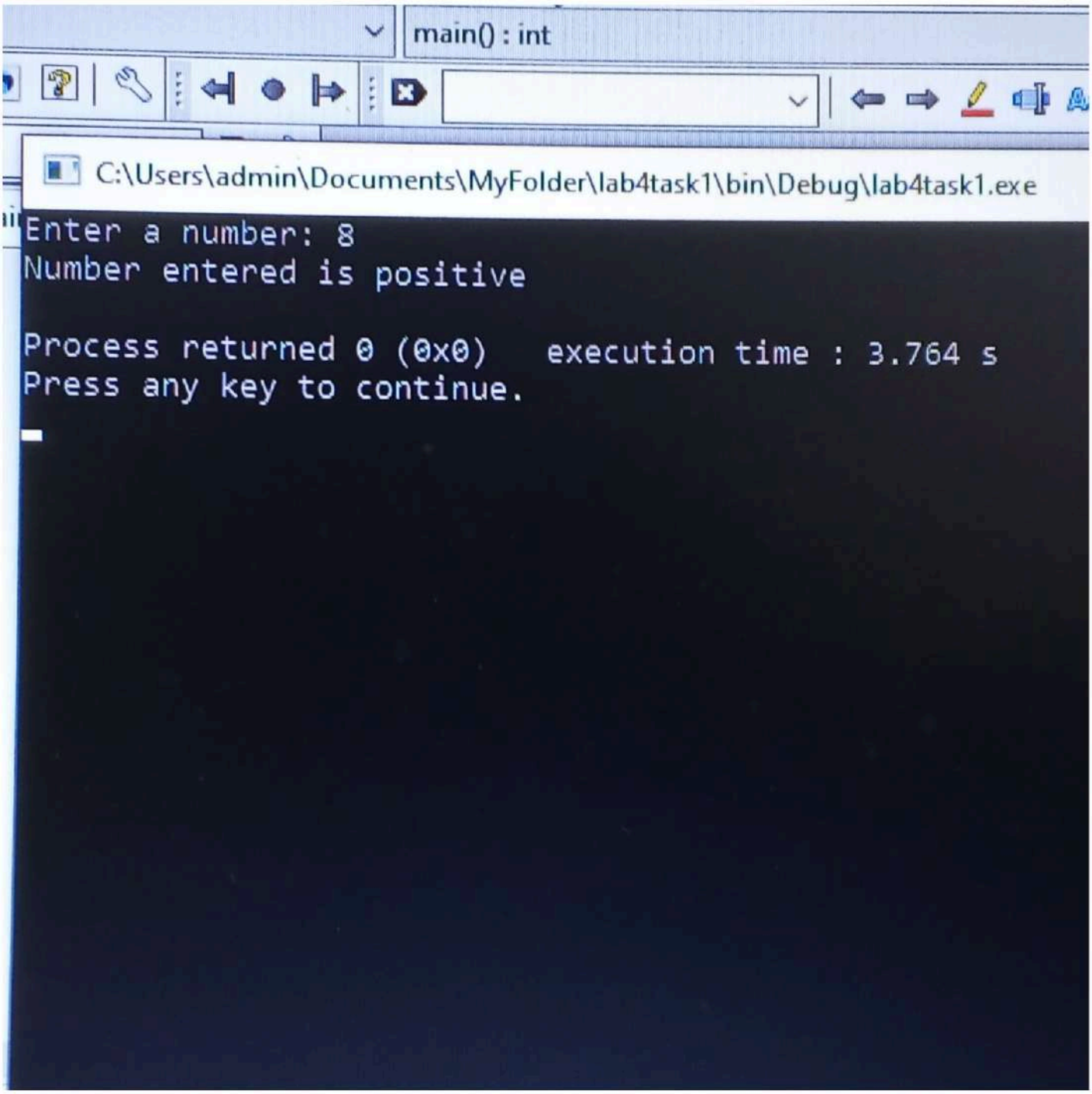
```
#include <iostream>
Using namespace std;
 int main () {
        int num ;
        cout << " Enter a number : ";
          cin >> num ;
    if (num >= 0) {
          cout <<" Number entered is positive "<<endl;
} else {
    cout << " Number is negative "<<endl;
}

    return 0;
}
```

Output

```
Enter a number : 8
Number entered is positive
```


Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings H
Debug
main() : int
*main.cpp
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5 int num;
6 cout << "Enter a number: ";
7 cin >> num;
8
9 if (num >= 0) {
10 cout << "Number entered is positive" << endl;
11 } else {
12 cout << "Number is negative" << endl;
13 }
14 return 0;
15 }
16
Logs & others
Code::Blocks
Search results
Cccc
Build log
Build messages

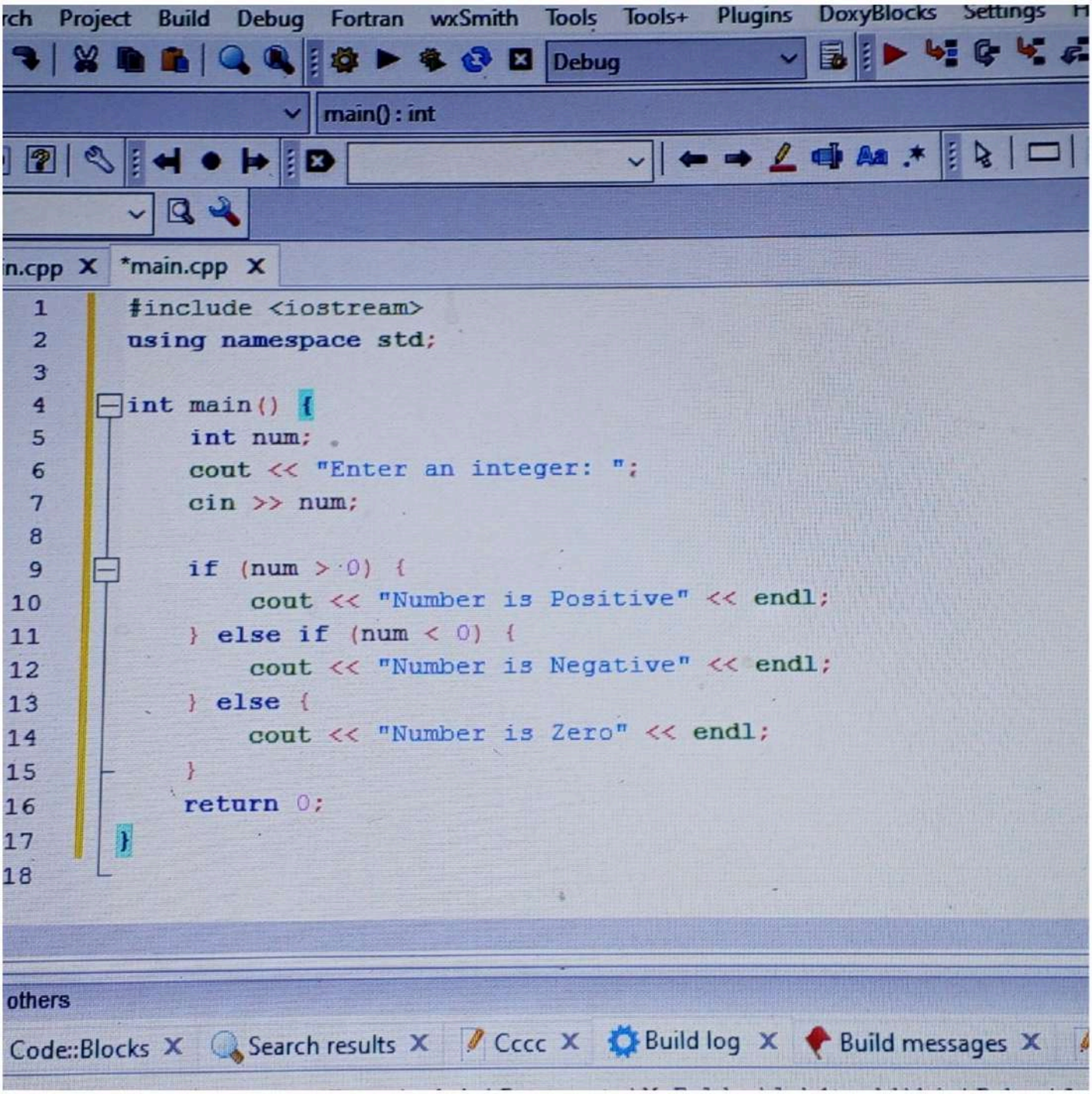

main() : int
C:\Users\admin\Documents\MyFolder\lab4task1\bin\Debug\lab4task1.exe
Enter a number: 8
Number entered is positive
Process returned 0 (0x0)   execution time : 3.764 s
Press any key to continue.

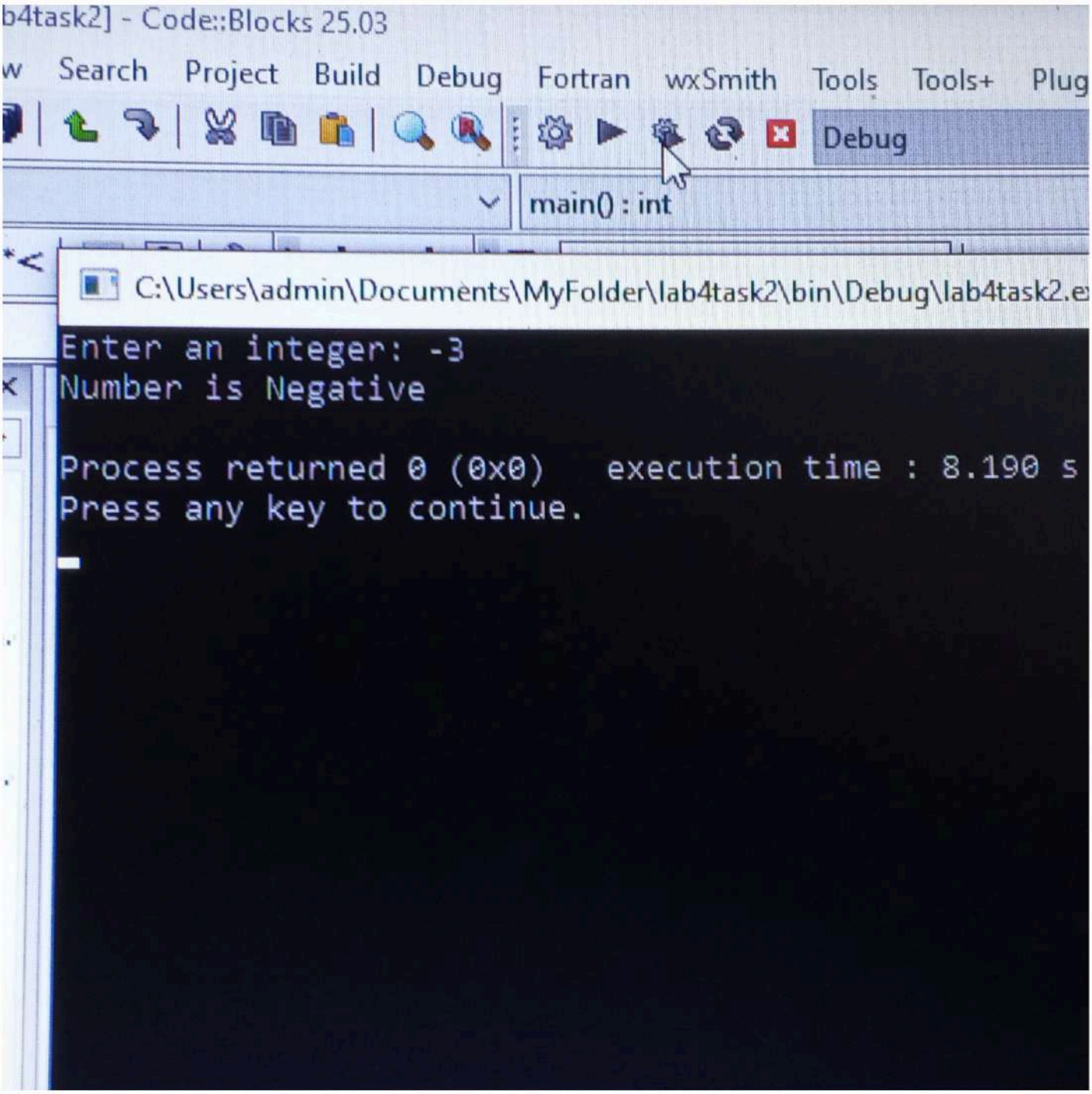
Lab-04 - Task 2

```
#include <iostream>
using namespace std;
int main ()
{
    int num;
    cout << " Enter an integer: ";
    cin >> num;
    if (num > 0) {
        cout << " Number is positive " << endl;
} else if (num < 0) {
    cout << " Number is Negative" << endl;
} else {
    cout << " Number is zero" << endl;
}
    return 0;
}
```

Output

```
Enter an integer :-3
Number is Negative
```


rch Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings H
Debug
main() : int
n.cpp X *main.cpp X
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5 int num;
6 cout << "Enter an integer: ";
7 cin >> num;
8
9 if (num > 0) {
10 cout << "Number is Positive" << endl;
11 } else if (num < 0) {
12 cout << "Number is Negative" << endl;
13 } else {
14 cout << "Number is Zero" << endl;
15 }
16 return 0;
17 }
18
others
Code::Blocks X Search results X Cccc X Build log X Build messages X


b4task2] - Code::Blocks 25.03
w Search Project Build Debug Fortran wxSmith Tools Tools+ Plug
Debug
main() : int
C:\Users\admin\Documents\MyFolder\lab4task2\bin\Debug\lab4task2.e
Enter an integer: -3
Number is Negative
Process returned 0 (0x0) execution time : 8.190 s
Press any key to continue.

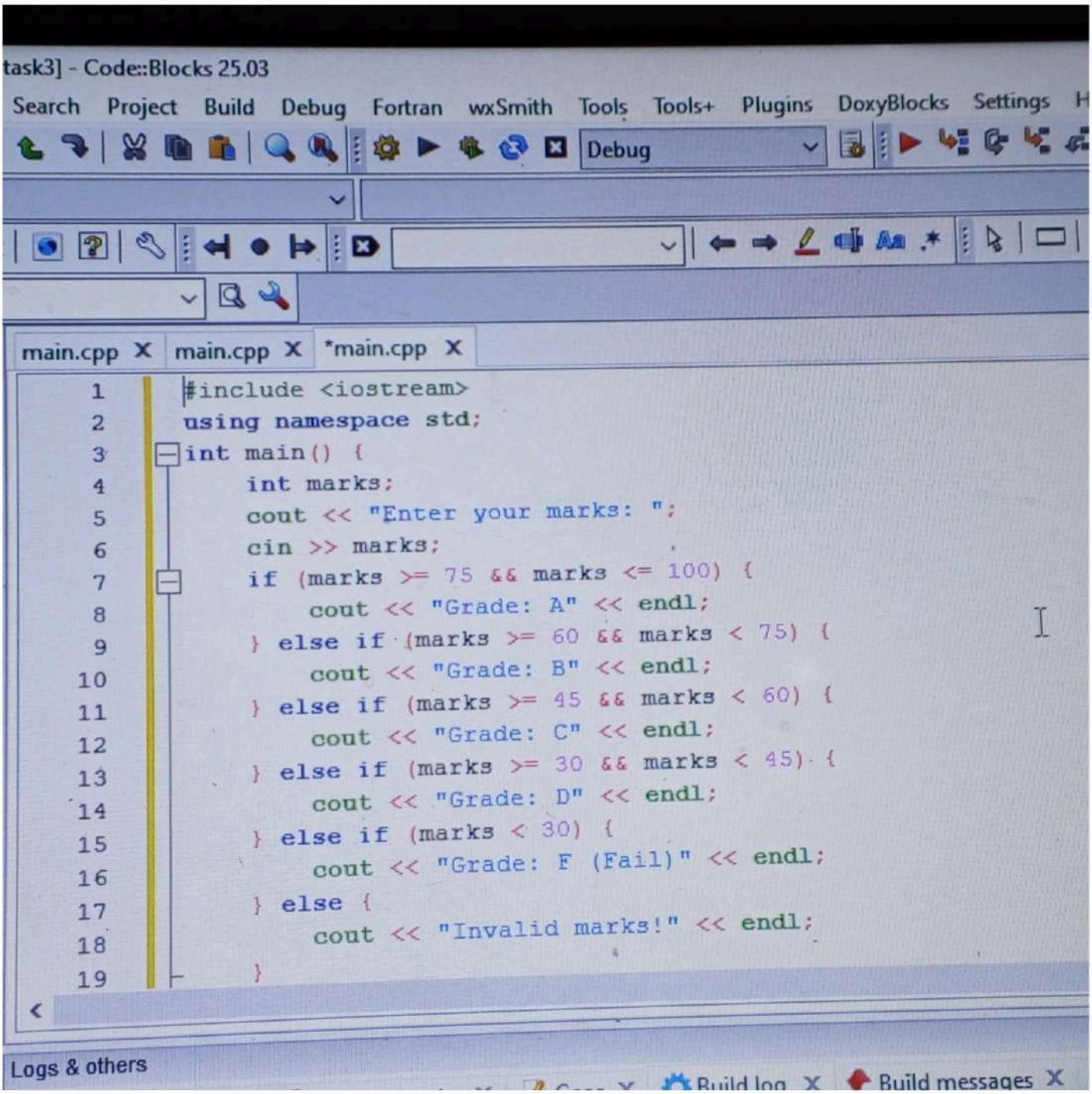
Lab-04 Task-03

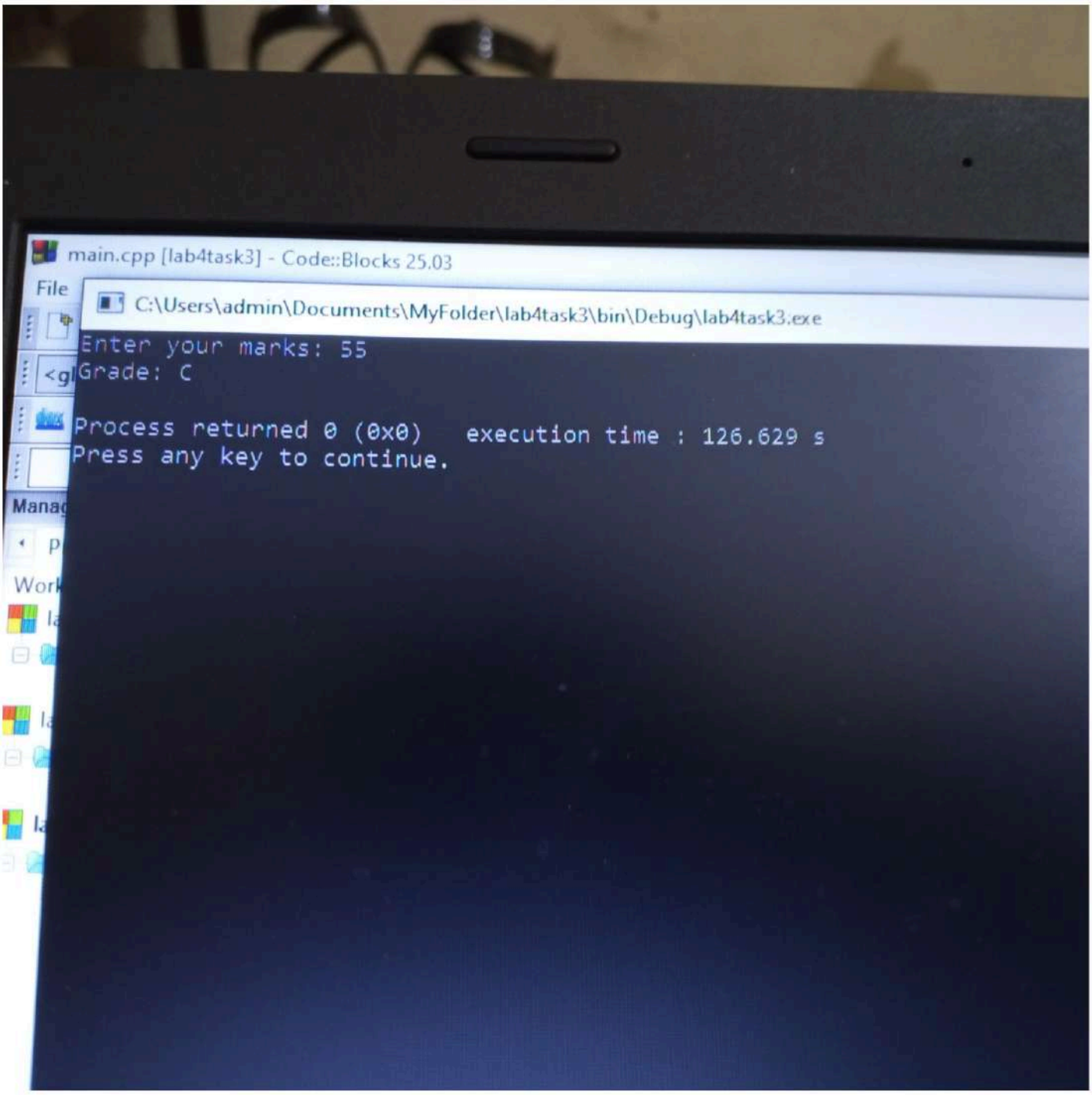
```
# include <iostream>
using namespace std;
int main ()
{
    int marks;
    cout << " Enter your marks:";
    cin >> marks;
   if (marks >= 75 && marks <=100) {
     cout << " Grade :A" <<endl;
} else if (marks >=60 && marks<75) {
     cout << " Grade: B" << endl;
} else if ( marks >=45 && marks<60) {
     cout << " Grade: C" << endl;
} else if (marks >= 30 && marks<45) {
     cout << " Grade:D" << endl;
 } else if (marks <30) {
    cout << " Grade : F(fail)" << endl;
 } else {
     cout << "Invalid marks!" << endl;
 }
  return 0;
}
```

output

Enter your marks: 55

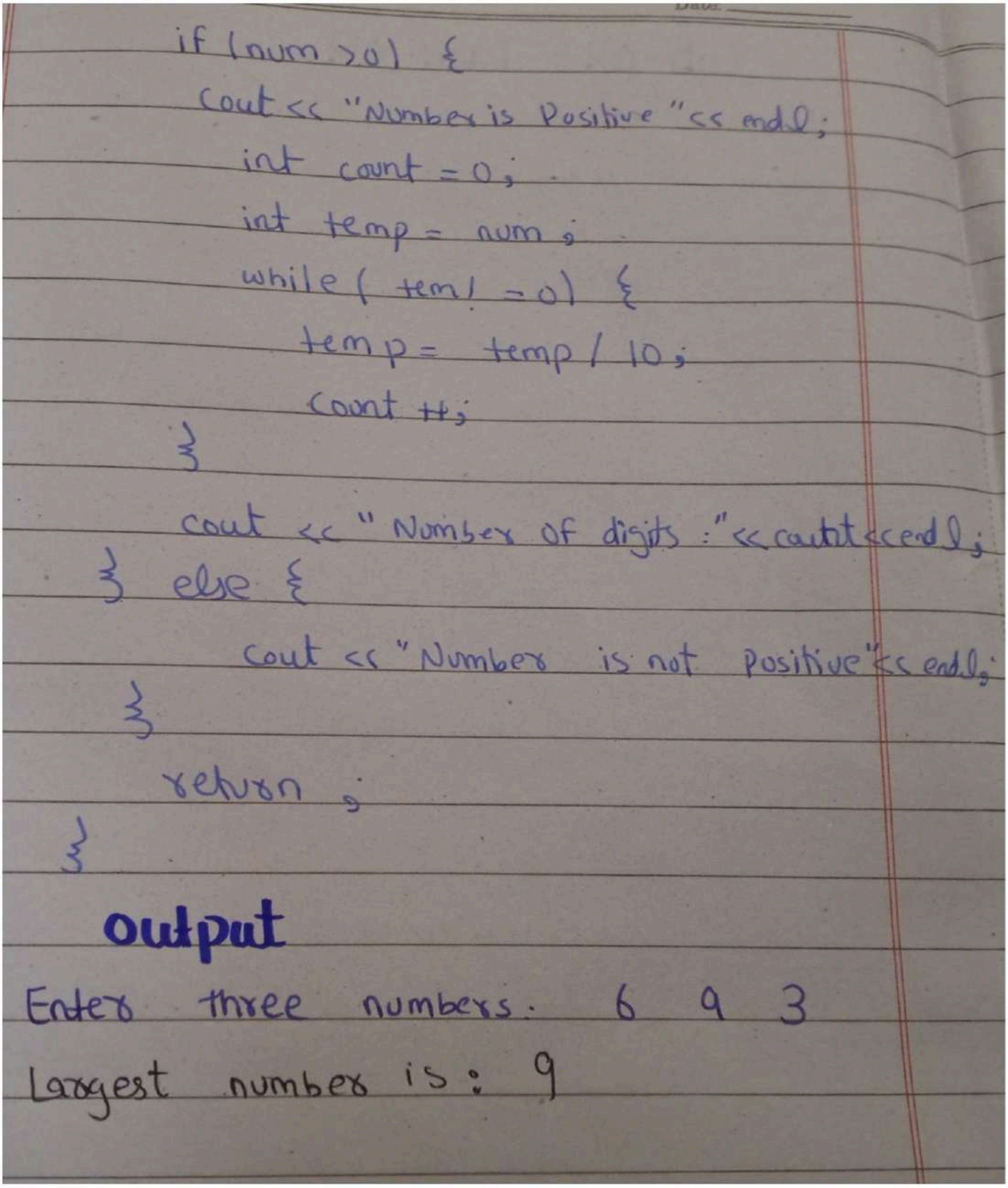
Grade: C


task3] - Code::Blocks 25.03
Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings
Debug
main.cpp
main.cpp
*main.cpp
#include <iostream>
using namespace std;
int main() {
int marks;
cout << "Enter your marks: ";
cin >> marks;
if (marks >= 75 && marks <= 100) {
cout << "Grade: A" << endl;
} else if (marks >= 60 && marks < 75) {
cout << "Grade: B" << endl;
} else if (marks >= 45 && marks < 60) {
cout << "Grade: C" << endl;
} else if (marks >= 30 && marks < 45) {
cout << "Grade: D" << endl;
} else if (marks < 30) {
cout << "Grade: F (Fail)" << endl;
} else {
cout << "Invalid marks!" << endl;
}
Logs & others
Build log
Build messages


main.cpp [lab4task3] - Code::Blocks 25.03
File
C:\Users\admin\Documents\MyFolder\lab4task3\bin\Debug\lab4task3.exe
Enter your marks: 55
Grade: C
Process returned 0 (0x0)   execution time : 126.629 s
Press any key to continue.

Lab-04 - Task. 04

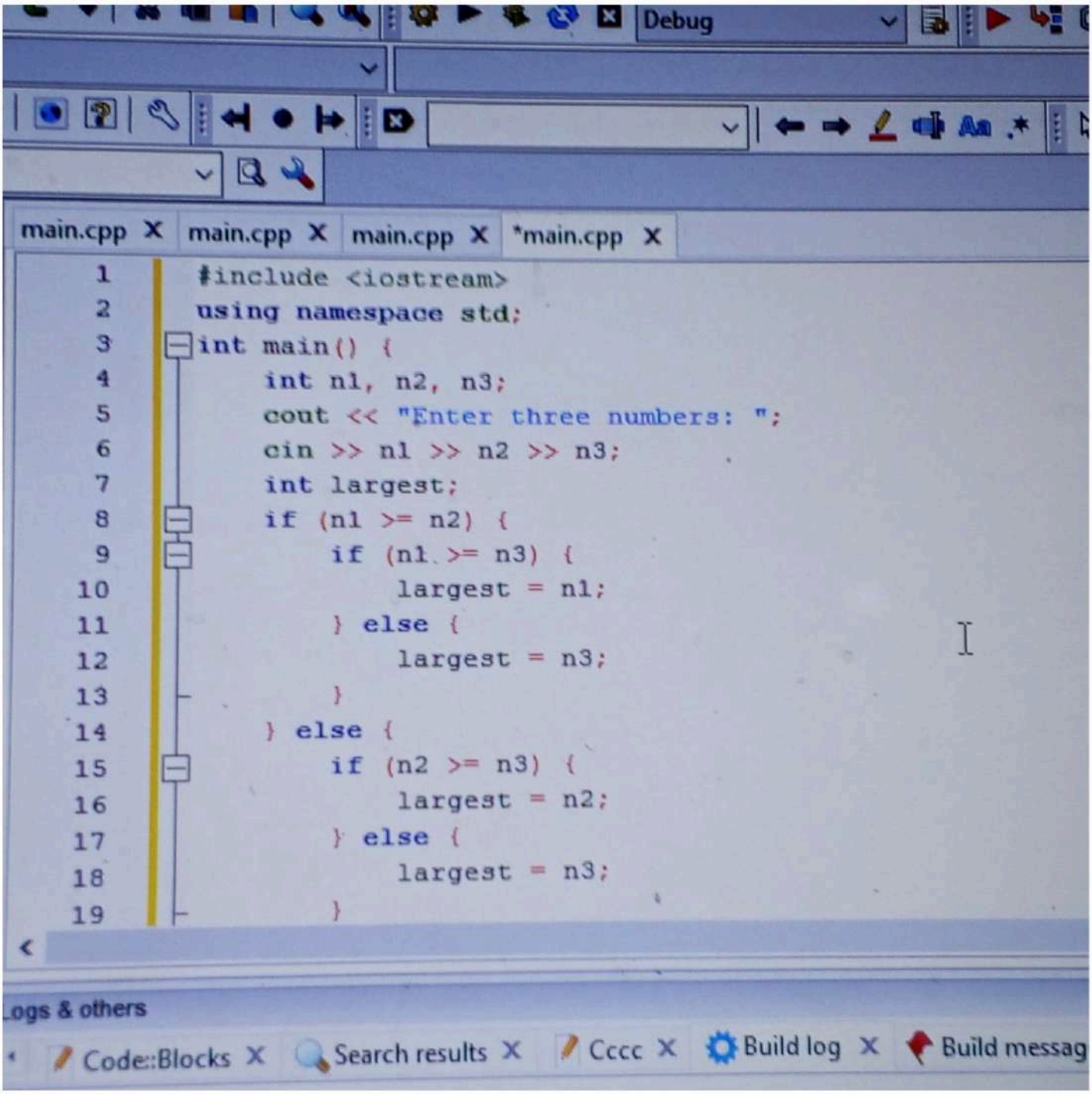
```
# include <iostream>
Using namespace std;
int main ( )  {
        int n1, n2, n3;
      cout<< "Enter Three numbers:" ;
       cin >> n1 >> n2>>n3 ;
       int largest;
       if(n1 >= n2) {
            if (n1 >= n3) {
            Largest =n1;
      } else {
            large  = n3;
        }
    } else {
        if (n2 >= n3) {
          largest = n2;
      } else {
            largest = n3;
      }
}
      Cout << " largest Number is:" << large<< endl;
      int num;
      cout << "\n Enter a number to check digits:";
          cin >> num;
```

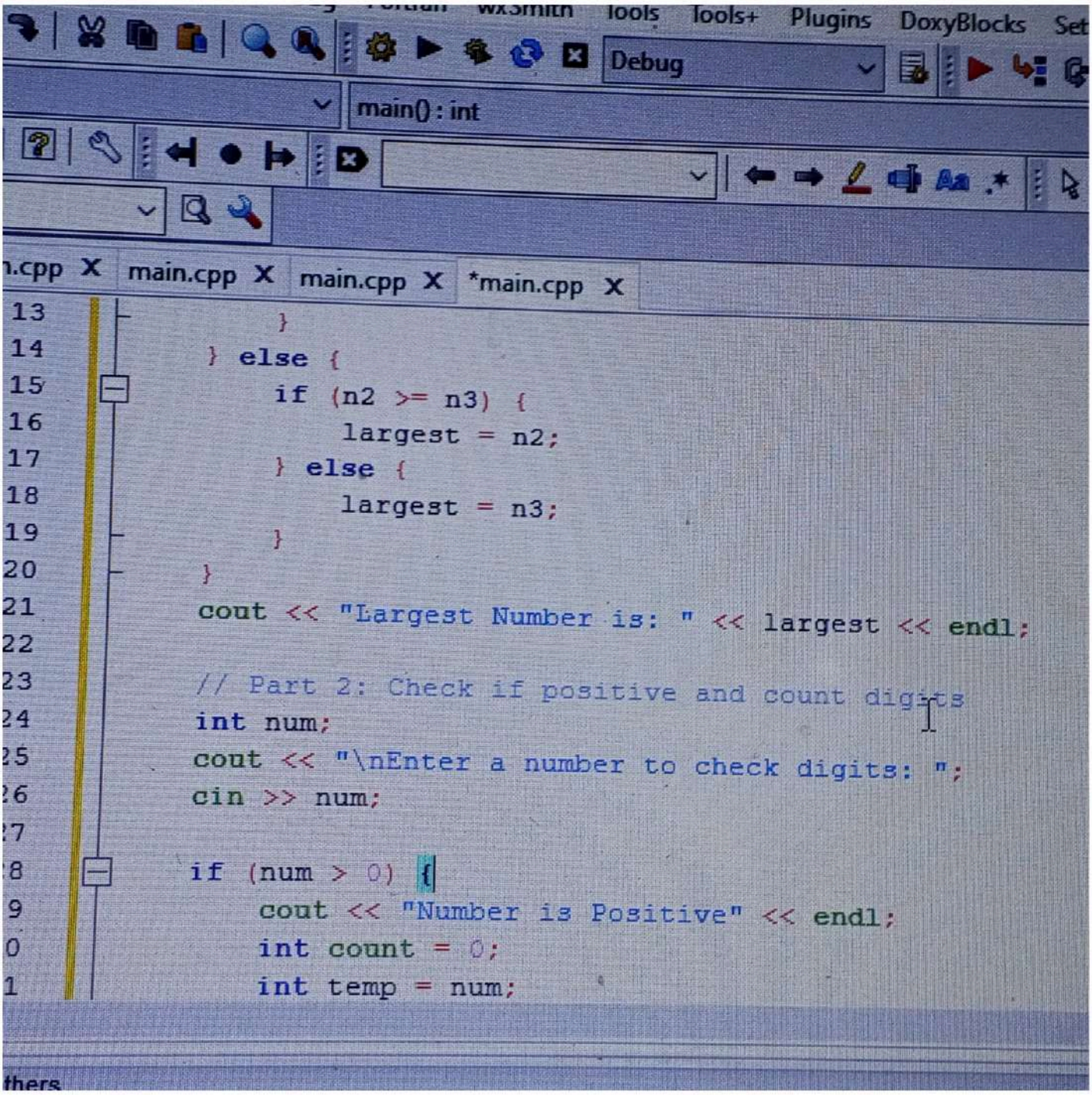


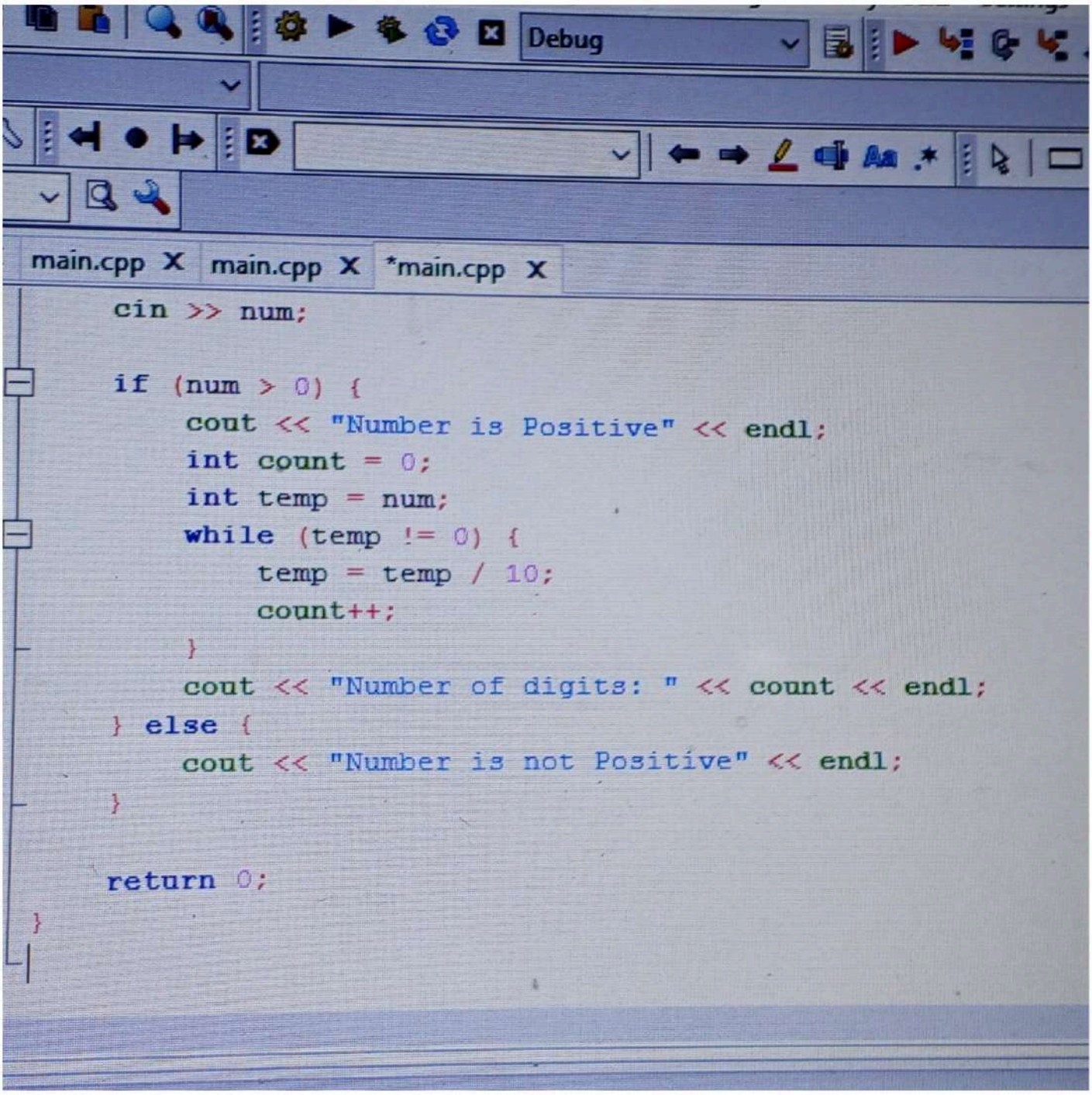
```
if (num >0) {
cout << "Number is Positive" << endl;
    int count = 0;
    int temp = num;
    while ( temp != 0) {
        temp = temp / 10;
        count ++;
    }
    cout << " Number of digits :" << count << endl;
} else {
    cout << "Number is not positive" << endl;
}
    return ;
}
```

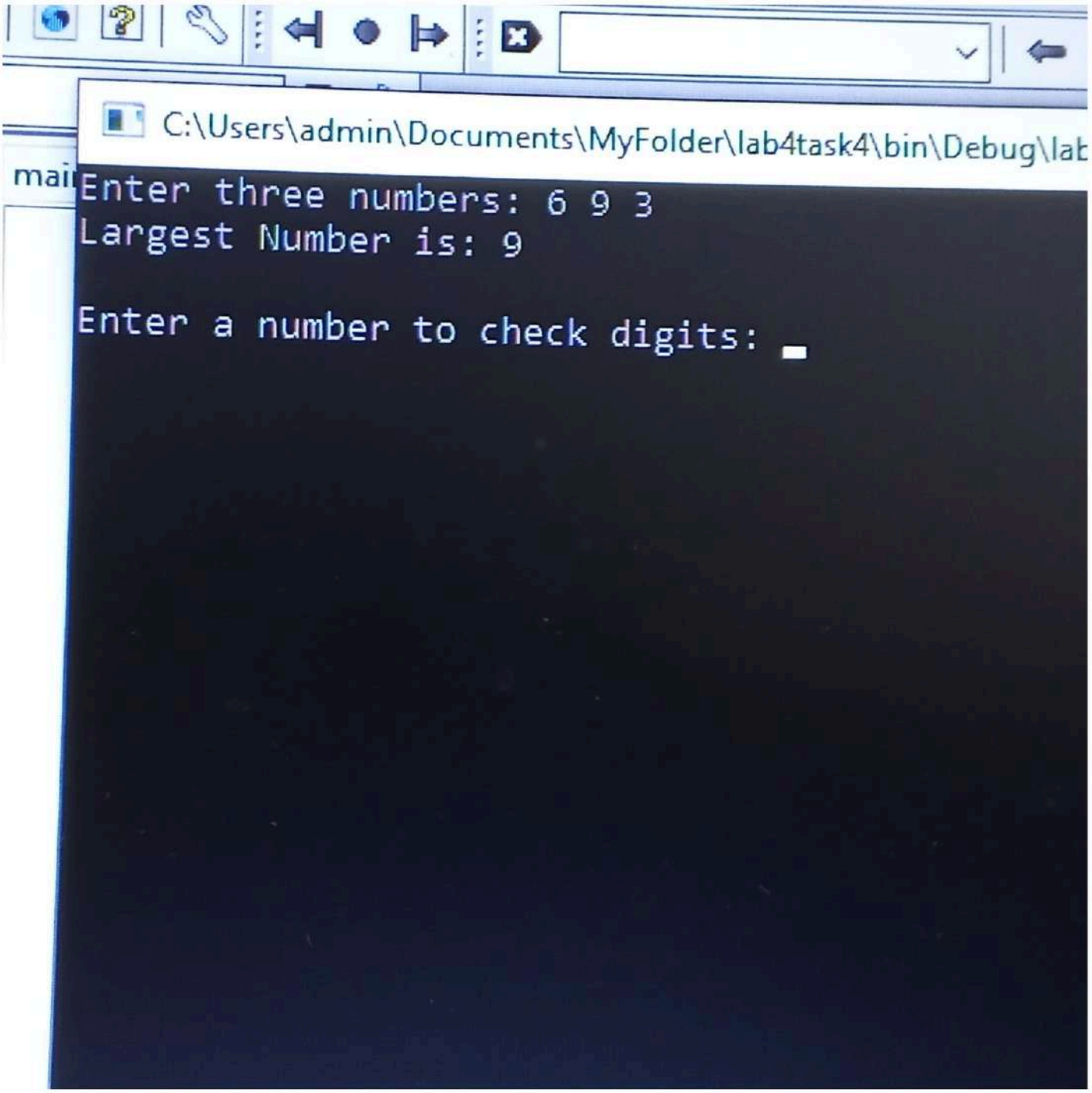
output

```
Enter three numbers: 6 9 3
Largest number is: 9
```


Debug
main.cpp
main.cpp
main.cpp
*main.cpp
#include <iostream>
using namespace std;
int main() {
int n1, n2, n3;
cout << "Enter three numbers: ";
cin >> n1 >> n2 >> n3;
int largest;
if (n1 >= n2) {
if (n1 >= n3) {
largest = n1;
} else {
largest = n3;
}
} else {
if (n2 >= n3) {
largest = n2;
} else {
largest = n3;
}
Logs & others
Code::Blocks
Search results
Cccc
Build log
Build messag


wxSmith
Tools
Tools+
Plugins
DoxyBlocks
Set
Debug
main() : int
n.cpp
main.cpp
main.cpp
*main.cpp
13 }
14 } else {
15 if (n2 >= n3) {
16 largest = n2;
17 } else {
18 largest = n3;
19 }
20 }
21 cout << "Largest Number is: " << largest << endl;
22
23 // Part 2: Check if positive and count digits
24 int num;
25 cout << "\nEnter a number to check digits: ";
26 cin >> num;
27
28 if (num > 0) {
29 cout << "Number is Positive" << endl;
30 int count = 0;
31 int temp = num;
thers


Debug
main.cpp
main.cpp
*main.cpp
cin >> num;
if (num > 0) {
cout << "Number is Positive" << endl;
int count = 0;
int temp = num;
while (temp != 0) {
temp = temp / 10;
count++;
}
cout << "Number of digits: " << count << endl;
} else {
cout << "Number is not Positive" << endl;
}
return 0;
}


C:\Users\admin\Documents\MyFolder\lab4task4\bin\Debug\lab
Enter three numbers: 6 9 3
Largest Number is: 9
Enter a number to check digits: